Team Name:6_EmpTrack

Domain: Human Resource and Staff Administration

Problem Statement:Attendance,Leave and Work Allocation

Functional Requirements:

Functional requirements describe what the system is supposed to do, including the services it provides and the actions it performs in response to user input .

A. Access & Role Management

● System provides registration, login, and logout functionality for Employees, Managers, and HR/Admin

● System allows profile management for Employees, Managers, and HR/Admin

● System distinguishes between Employee, Manager, and HR/Admin roles

● System enforces role-based access control

B. Leave Management

● Employees can apply for leave by selecting leave type, dates, and reason

● System validates leave requests based on available leave balance

● Managers/HR can view pending leave requests

● Managers/HR can approve or reject leave requests

● Employees receive notifications for leave approval or rejection

● System updates leave balance automatically after approval

● Leave history is maintained for each employee

C. Work Allocation & Task Management

● Managers can create and assign work tasks to employees

● Tasks include description, priority, and deadline

● Employees can view assigned tasks

● Employees can update task status (pending, in-progress, completed)

● Managers can monitor task progress

● System maintains records of completed and pending tasks

D. Attendance Management

● Employees can mark daily attendance (check-in and check-out)

● System records date and time of attendance

● Duplicate attendance entries for the same day are prevented

● HR/Managers can view employee attendance records

● Monthly attendance reports can be generated

● Attendance data is linked with leave records

E. Reporting & Monitoring

● Generate reports for attendance, leave, and work status

● View employee-wise and department-wise summaries

● Audit logs maintained for system activities

Non-Functional Requirements:

Non-Functional requirements specify the quality attributes and constraints of the system such as performance, security, reliability, and usability.

A. Product Requirements

1. Efficiency Requirements

● The system should support multiple employees accessing the system simultaneously

● Leave, attendance, and work allocation operations should be processed efficiently

2. Performance Requirements

● Page response time should be less than or equal to 2 seconds

● Attendance marking and leave submission should be processed in real time

● Reports should be generated without noticeable delay

3. Space Requirements

● Efficient database storage should be used for employee, leave, and attendance records

● System should optimize storage for logs and reports

4. Usability Requirements

● System should have a simple and user-friendly interface

● Interface should be easy to use for employees, managers, and HR

● System should be accessible through web browsers

5. Dependability Requirements

● System should be available during working hours without failure

● Data should be backed up regularly

● System should recover quickly from failures

6. Security Requirements

● Secure login and authentication mechanisms must be provided

● Role-based access control should be enforced

● Sensitive data such as personal details and attendance records should be protected

B. Organizational Requirements

1. Operational Requirements

● System should operate on web-based platforms

● System should support common operating systems and browsers

2. Environmental Requirements

● System should function properly under normal network conditions

● System should support use in office environments

3. Development Requirements

● System should be developed using standard programming frameworks

● Modular design should be followed for easy maintenance

● System should be scalable for future enhancements

C. External Requirements

1. Regulatory Requirements

- System should comply with organizational policies
- Employee data handling should follow data protection guidelines

2. Ethical Requirements

- Employee data should not be misused
- Attendance and leave data should be handled fairly

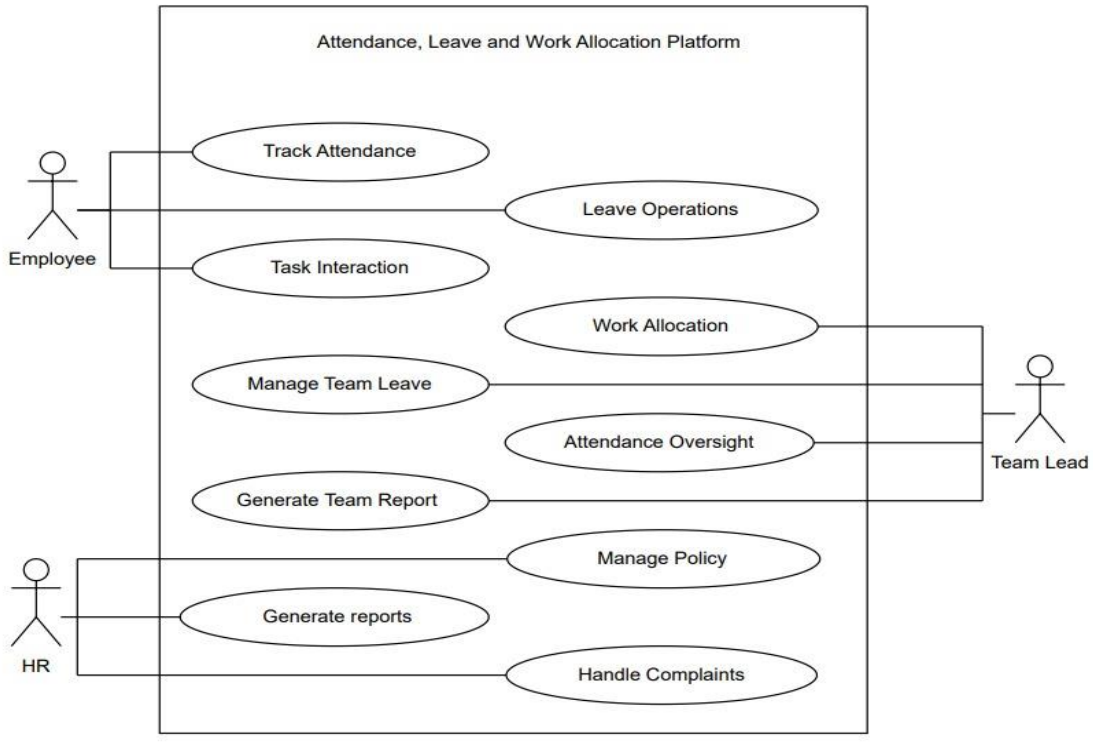
3. Legislative Requirements

- System should comply with IT laws and data privacy regulations
- Proper logs should be maintained for legal auditing

4. Safety / Security Requirements

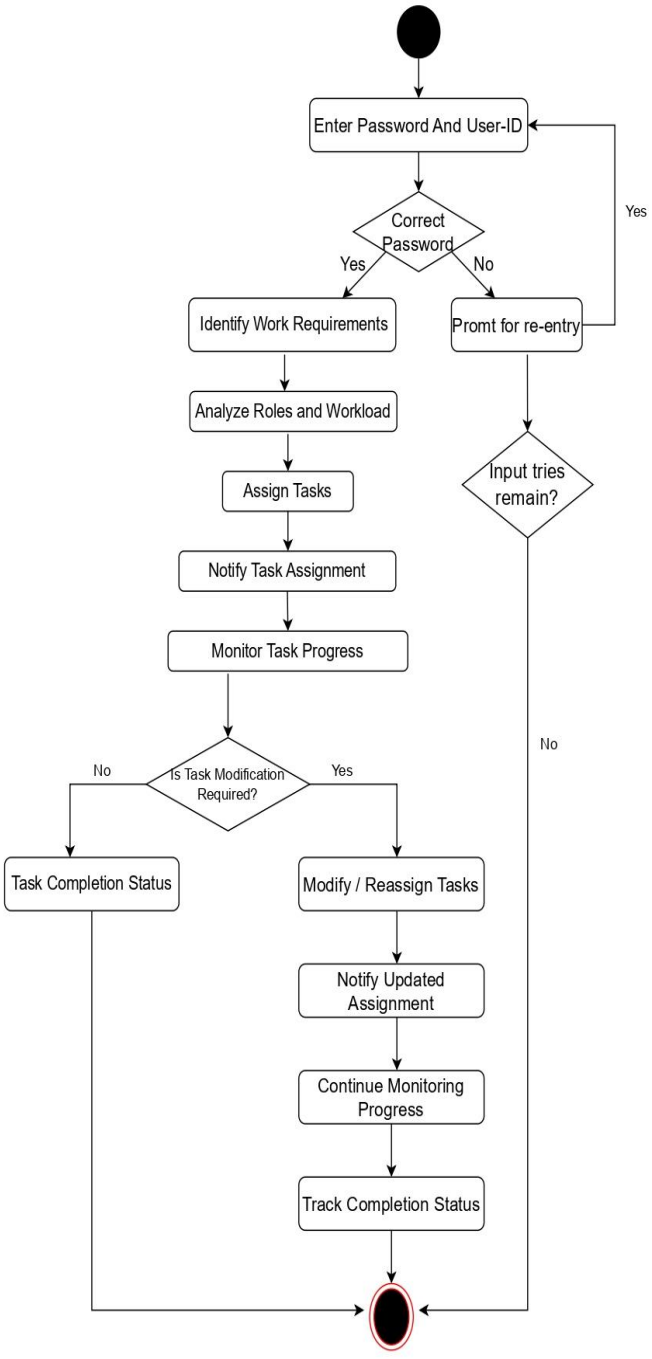
- System should be protected from unauthorized access
- Measures should be taken to prevent data loss or misuse

Use Case Diagram:

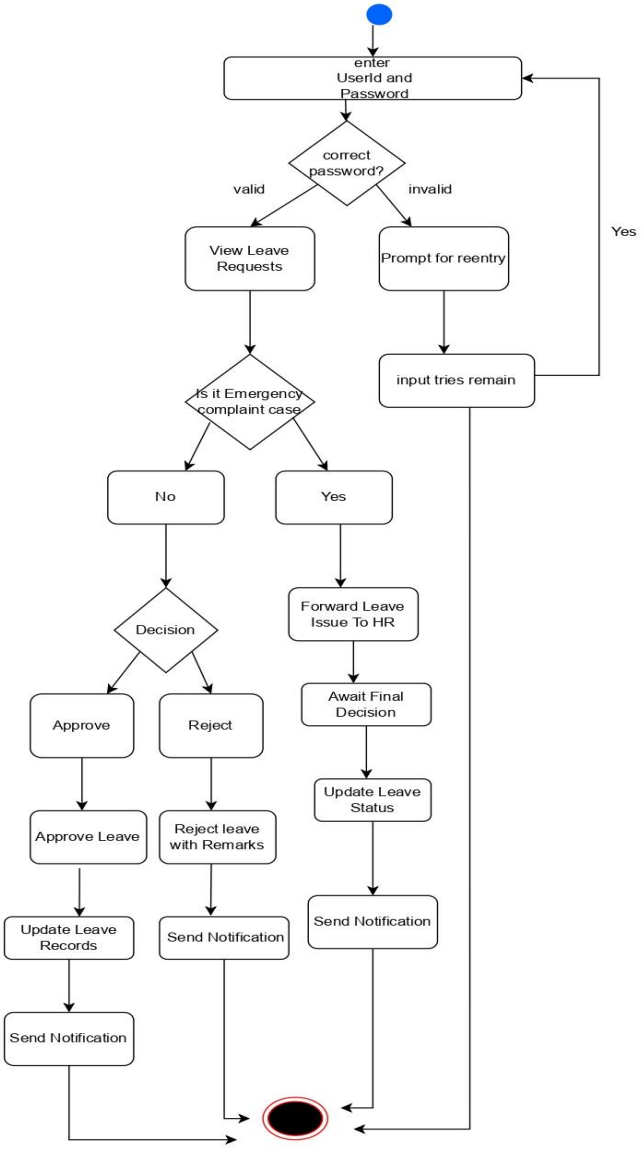


Activity Diagrams:

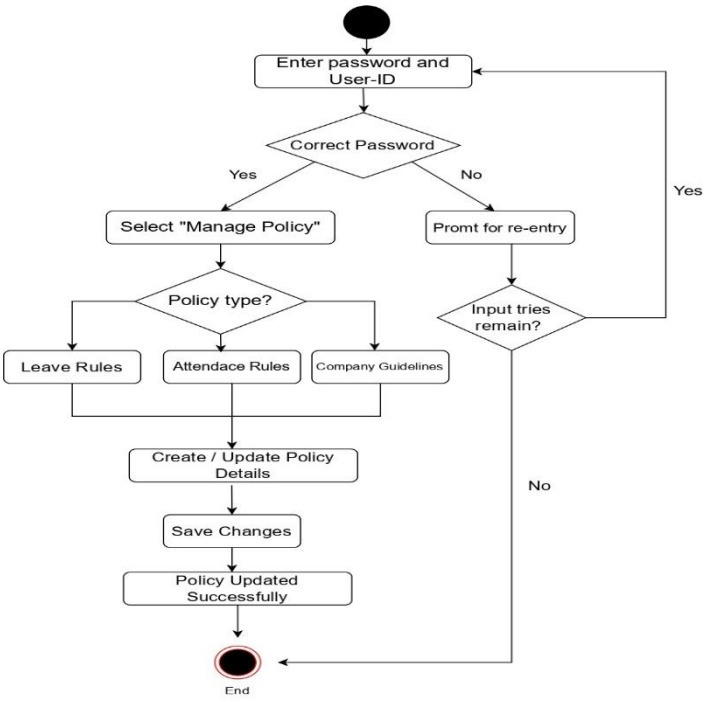
Assign Task Activity:



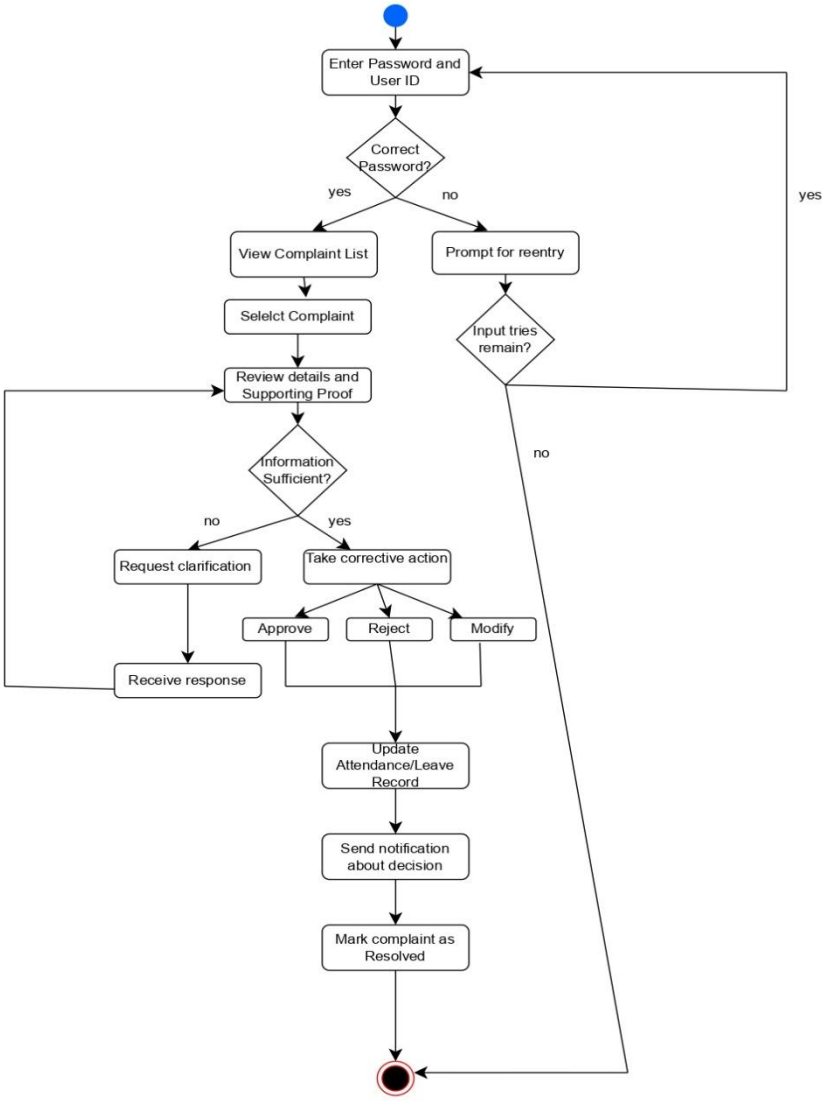
Manage Team Leave:



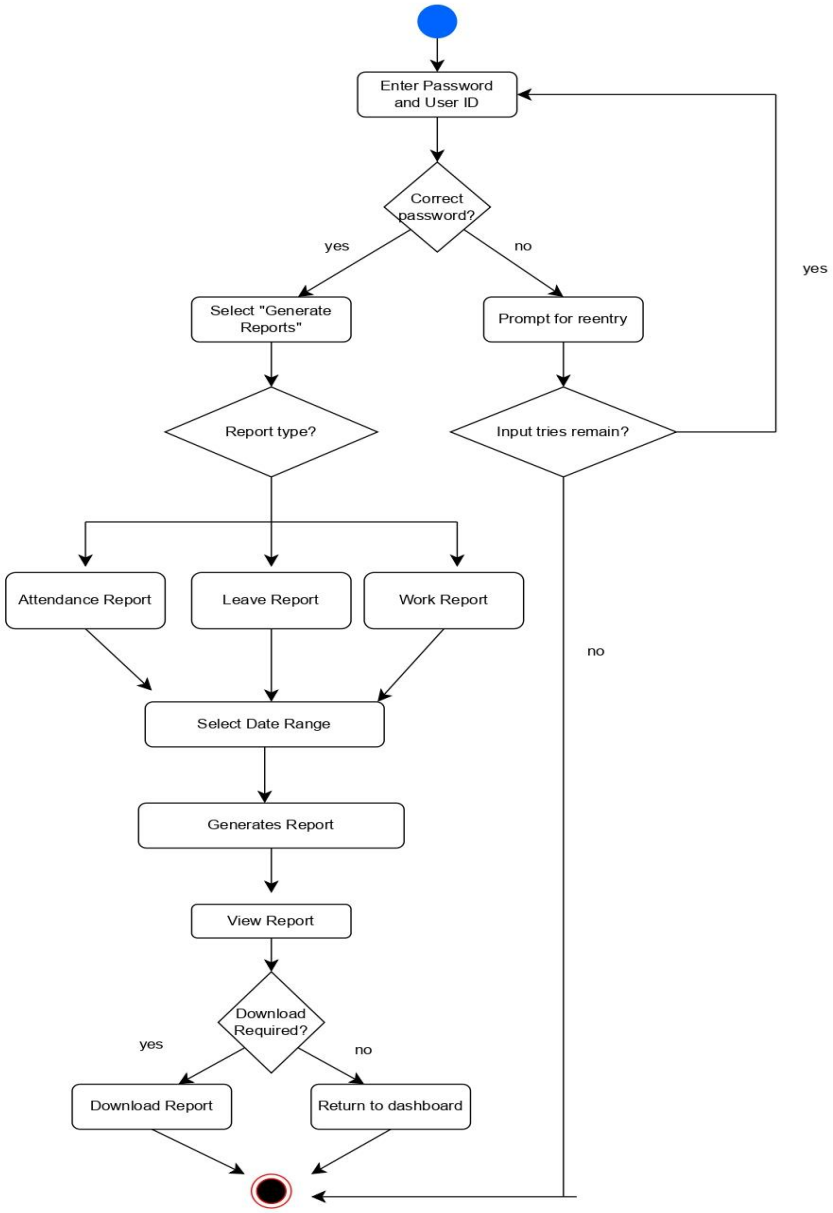
Manage Policy:



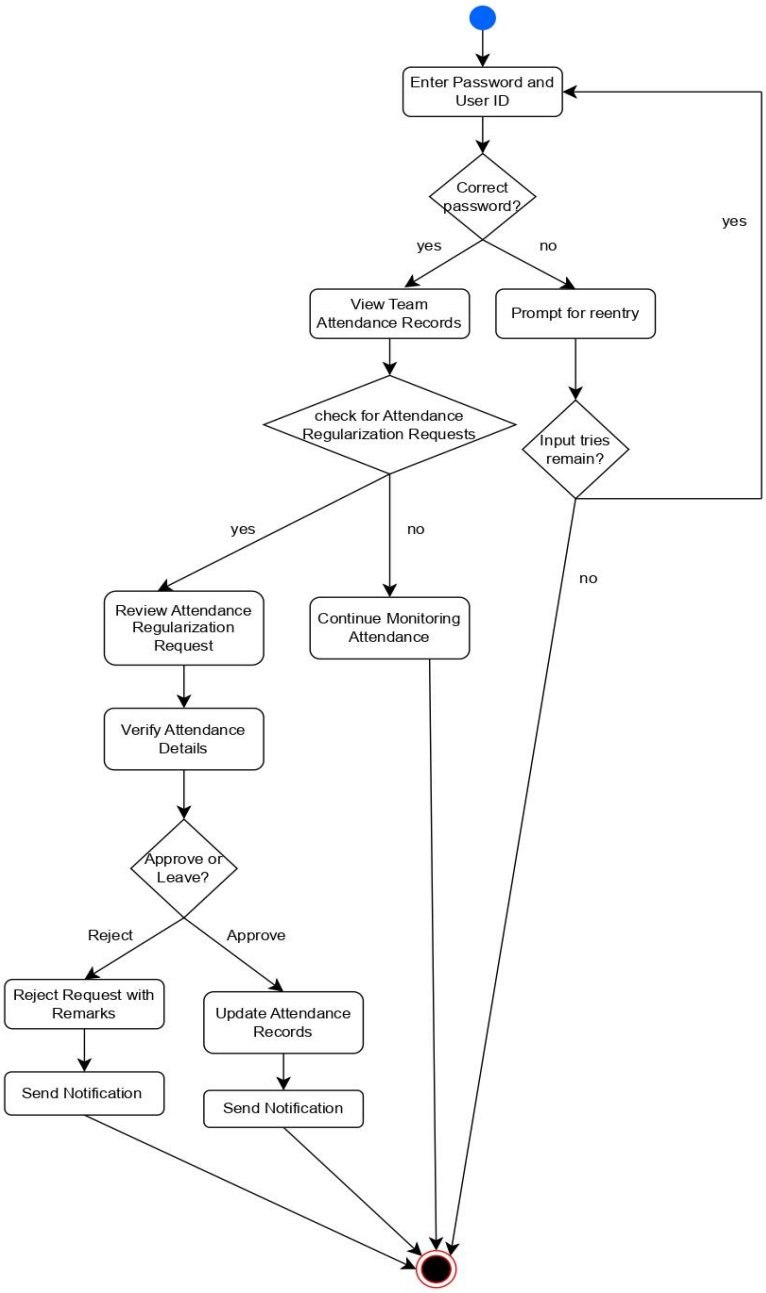
Handle complaints



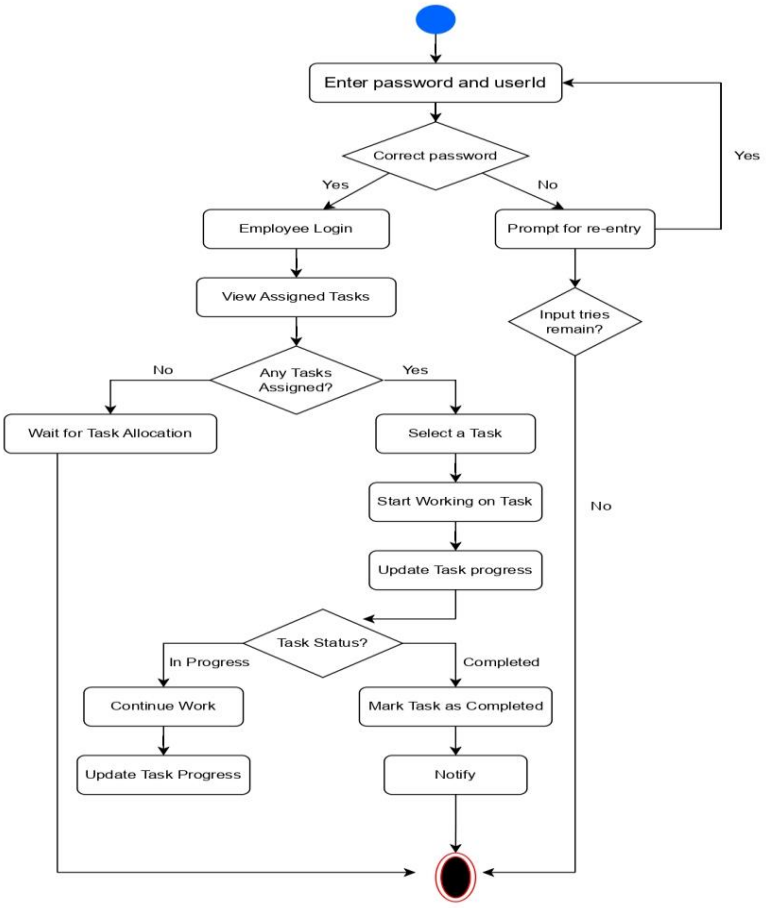
Generate Report



Manage Team Attendance

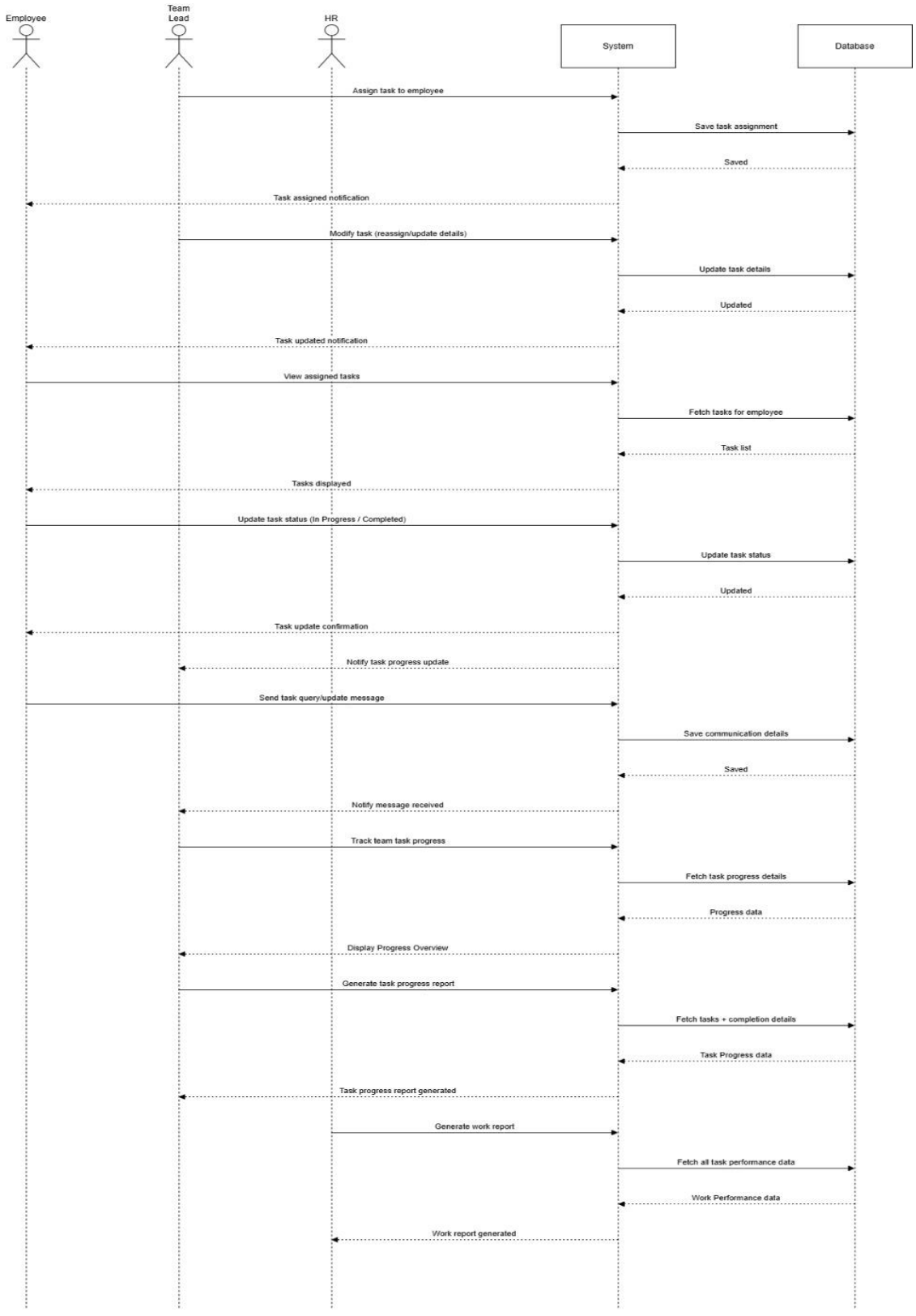


Task Interaction



Sequence Diagram

Work Allocation



Attendance and Leave Sequence Diagram:

